

IBS: Bezpečnost a počítačové sítě

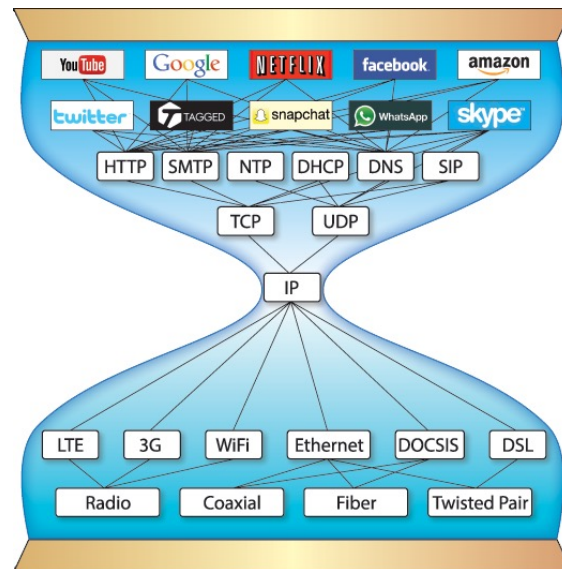
Network stack, containers basic

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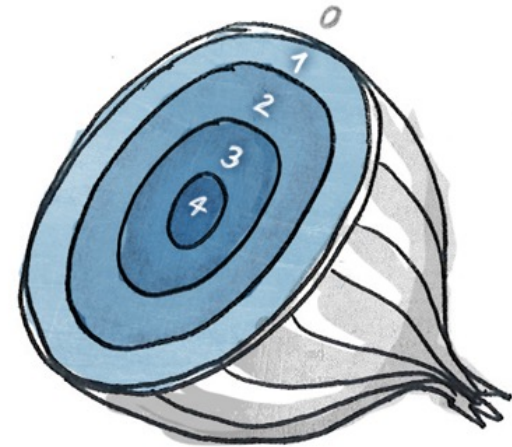
6th February 2025

Network Stack



- Computer networks are complex systems
- Too many “parts”
 - End-systems
 - Routers and switches
 - Cable systems
 - Applications
 - Protocol
 - HW and SW
- Thus, division of scope into layers
- Referential layered models
 - ISO/OSI
 - TCP/IP

- Modular layers
 - Relationship between subsystems
 - Transparent change of layer due to the well-defined APIs
 - Extensible communication

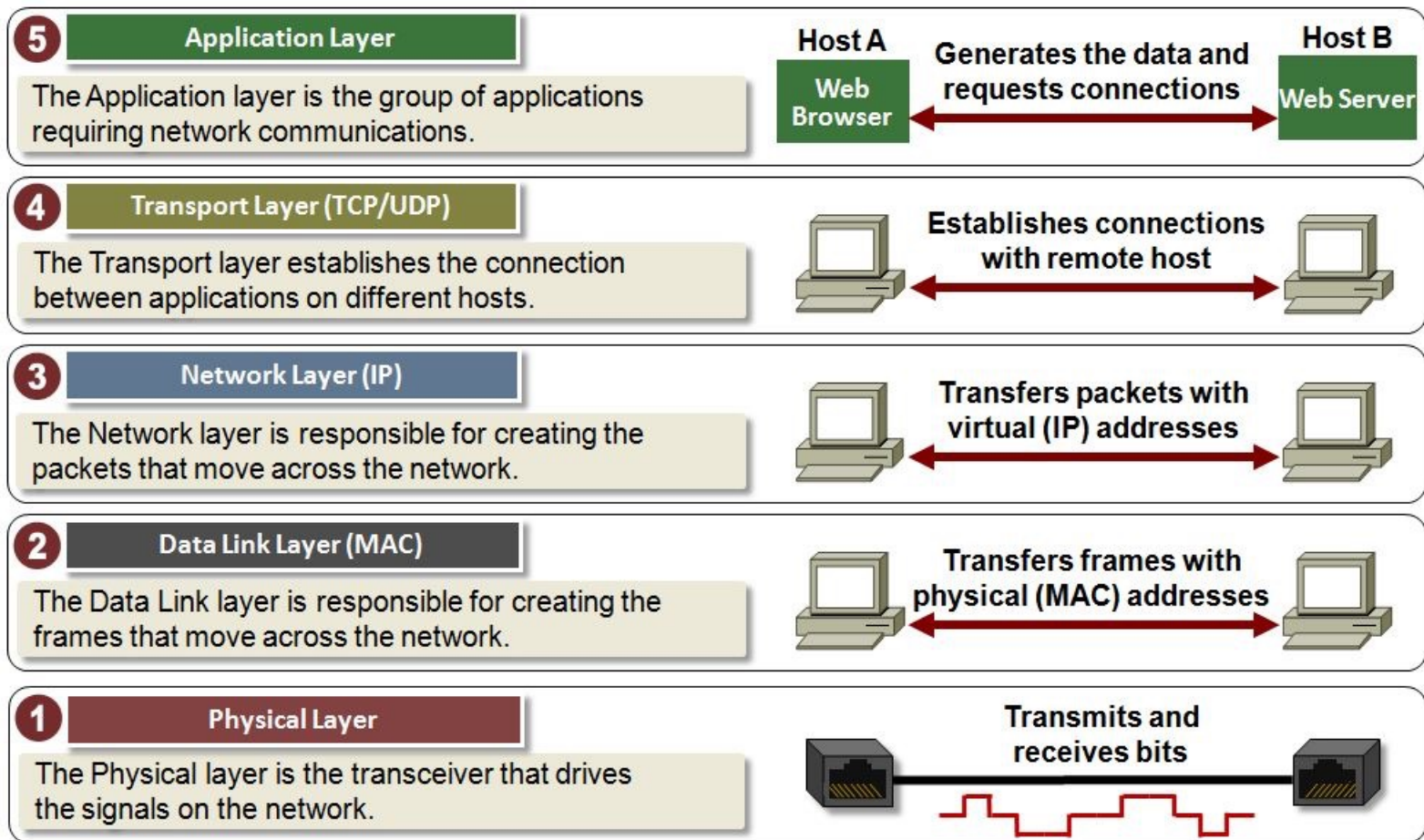




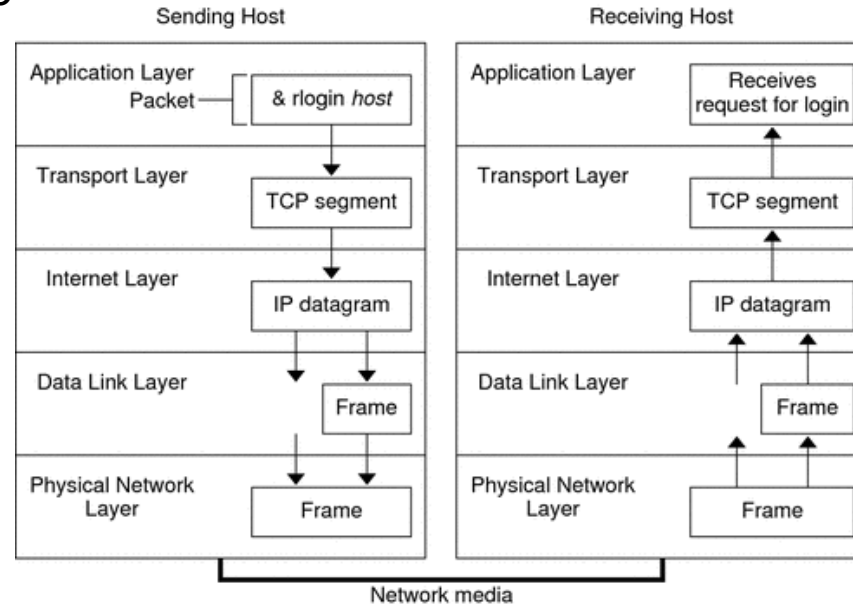
OSI MODEL		UPPER LAYERS
7	Application Layer Type of communication: E-mail, file transfer, client/server.	
6	Presentation Layer Encryption, data conversion: ASCII to EBCDIC, BCD to binary, etc.	
5	Session Layer Starts, stops session. Maintains order.	
4	Transport Layer Ensures delivery of entire file or message.	
3	Network Layer Routes data to different LANs and WANs based on network address.	LOWER LAYERS
2	Data Link (MAC) Layer Transmits packets from node to node based on station address.	
1	Physical Layer Electrical signals and cabling.	

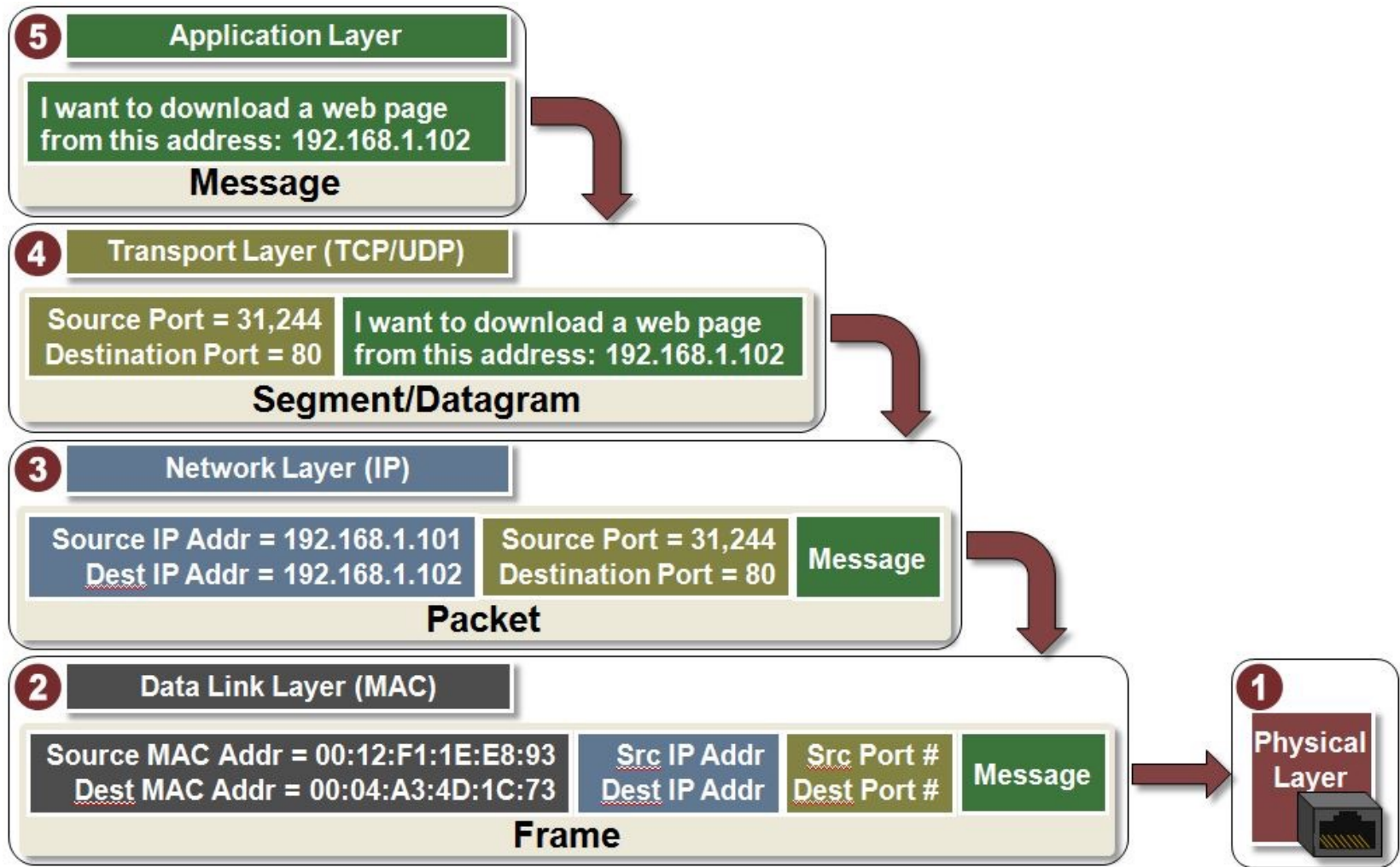
OSI (Open Source Interconnection) 7 Layer Model

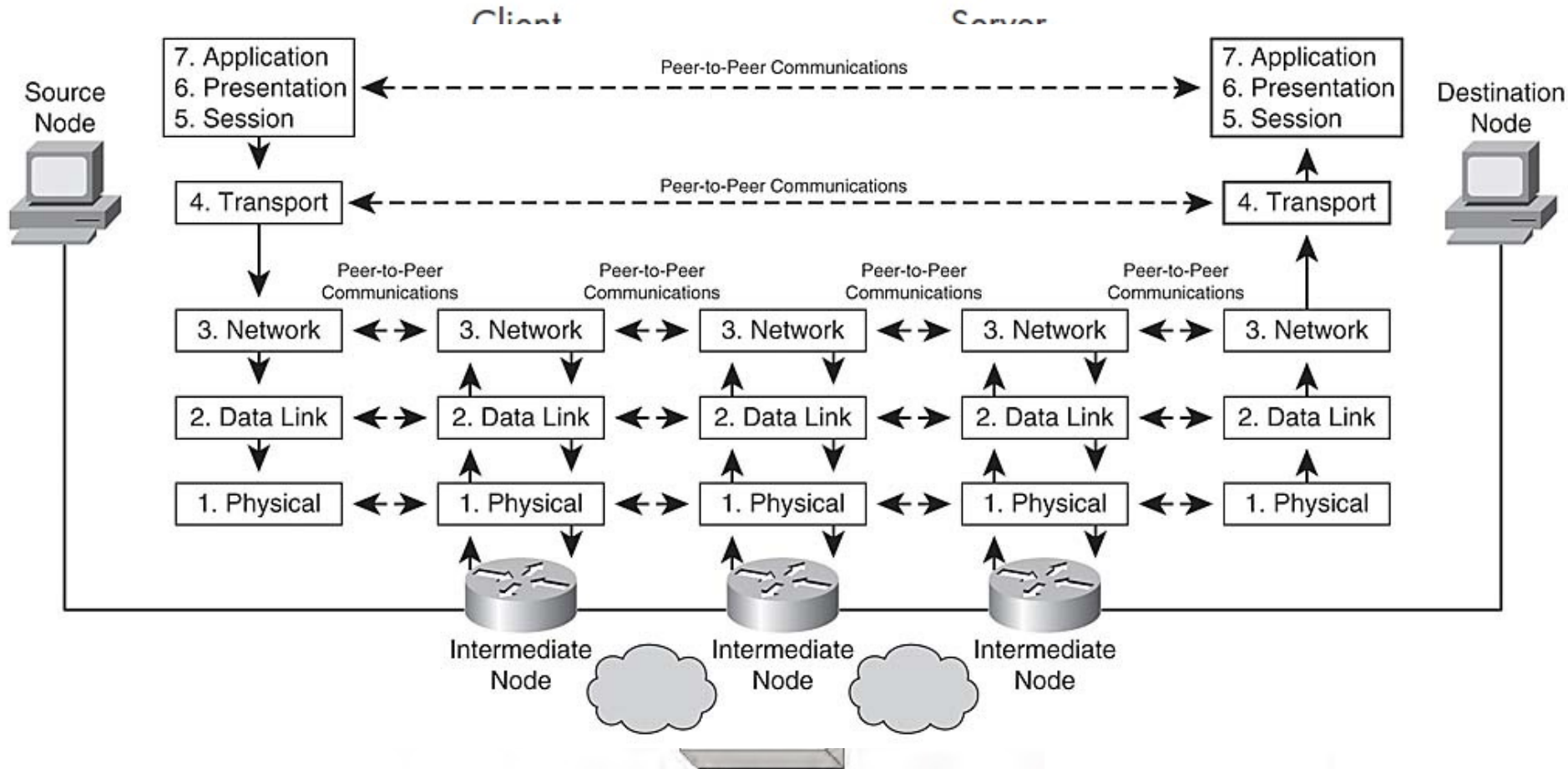
Layer	Application/Example		Central Device/Protocols		DOD4 Model
Application (7) Serves as the window for users and application processes to access the network services.	End User layer Program that opens what was sent or creates what is to be sent Resource sharing • Remote file access • Remote printer access • Directory services • Network management		User Applications SMTP	G A T E W A Y	Process
Presentation (6) Formats the data to be presented to the Application layer. It can be viewed as the "Translator" for the network.	Syntax layer encrypt & decrypt (if needed) Character code translation • Data conversion • Data compression • Data encryption • Character Set Translation		JPEG/ASCII EBDIC/TIFF/GIF PICT		
Session (5) Allows session establishment between processes running on different stations.	Synch & send to ports (logical ports) Session establishment, maintenance and termination • Session support - perform security, name recognition, logging, etc.		Logical Ports RPC/SQL/NFS NetBIOS names		
Transport (4) Ensures that messages are delivered error-free, in sequence, and with no losses or duplications.	TCP Host to Host, Flow Control Message segmentation • Message acknowledgement • Message traffic control • Session multiplexing	F I L T E R I N G P A C K E T	TCP/SPX/UDP		Host to Host
Network (3) Controls the operations of the subnet, deciding which physical path the data takes.	Packets ("letter", contains IP address) Routing • Subnet traffic control • Frame fragmentation • Logical-physical address mapping • Subnet usage accounting		Routers IP/IPX/ICMP		Internet
Data Link (2) Provides error-free transfer of data frames from one node to another over the Physical layer.	Frames ("envelopes", contains MAC address) [NIC card — Switch — NIC card] (end to end) Establishes & terminates the logical link between nodes • Frame traffic control • Frame sequencing • Frame acknowledgment • Frame delimiting • Frame error checking • Media access control	Switch Bridge WAP PPP/SLIP	Land Based Layers	Network	
Physical (1) Concerned with the transmission and reception of the unstructured raw bit stream over the physical medium.	Physical structure Cables, hubs, etc. Data Encoding • Physical medium attachment • Transmission technique - Baseband or Broadband • Physical medium transmission Bits & Volts	Hub			

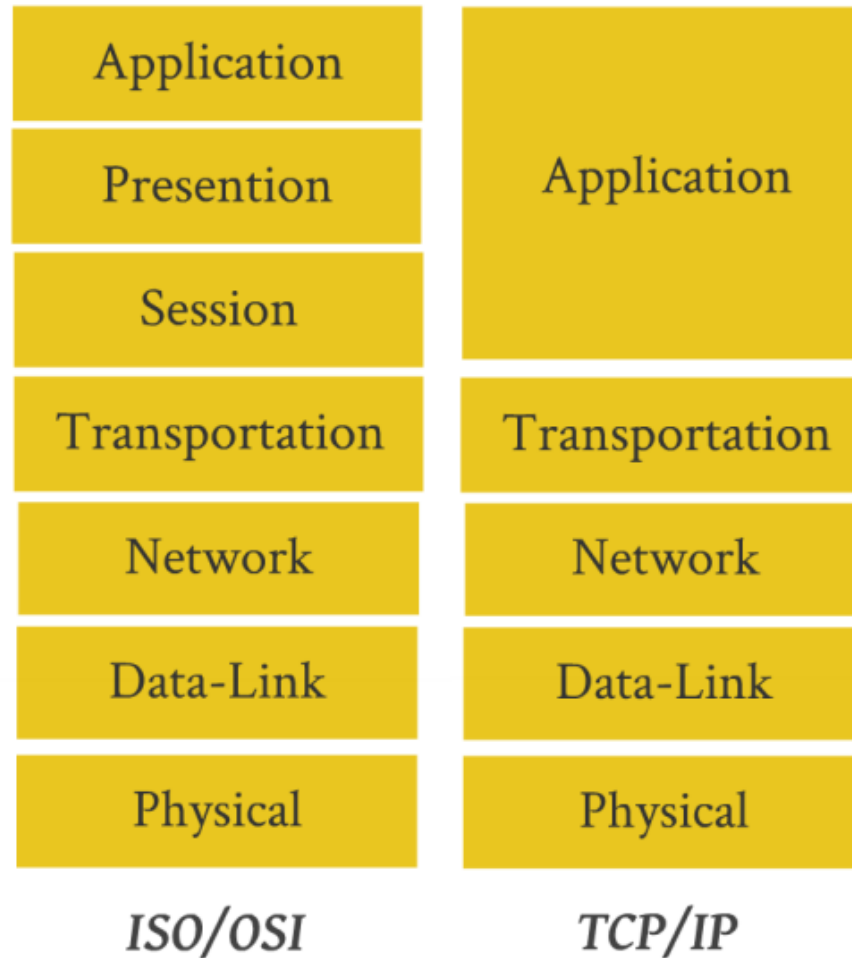


- Unit of information processed by a given layer
 - PDU consists of header + payload + (optionally) trailer
 - As data traverses through layers
 - down: encapsulation occurs, header is appended
 - up: decapsulation occurs, header is stripped
- PDU taxonomy
 - L7 PDU = application data
 - L4 PDU = **segments** (TCP) **datagrams** (UDP)
 - L3 PDU = **packets**
 - L2 PDU = **frames**
 - L1 PDU = bits

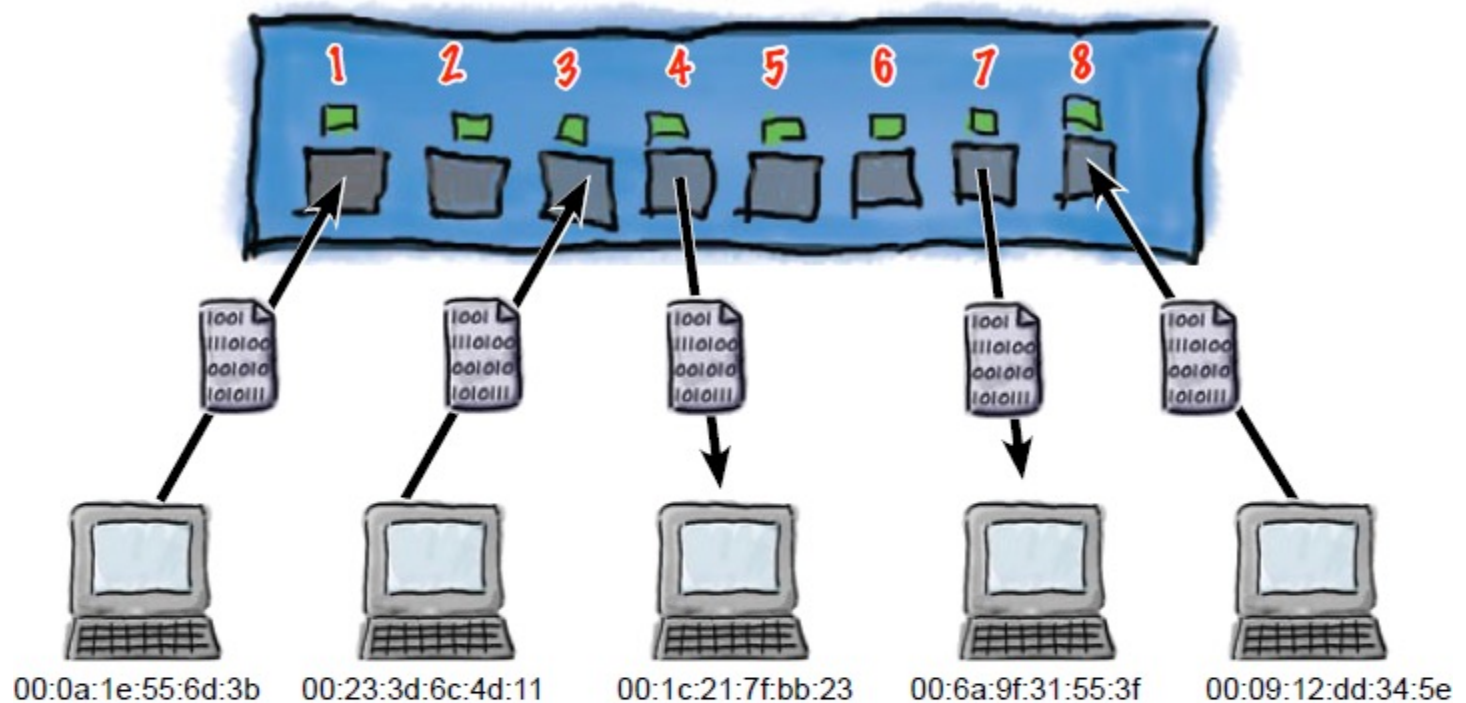




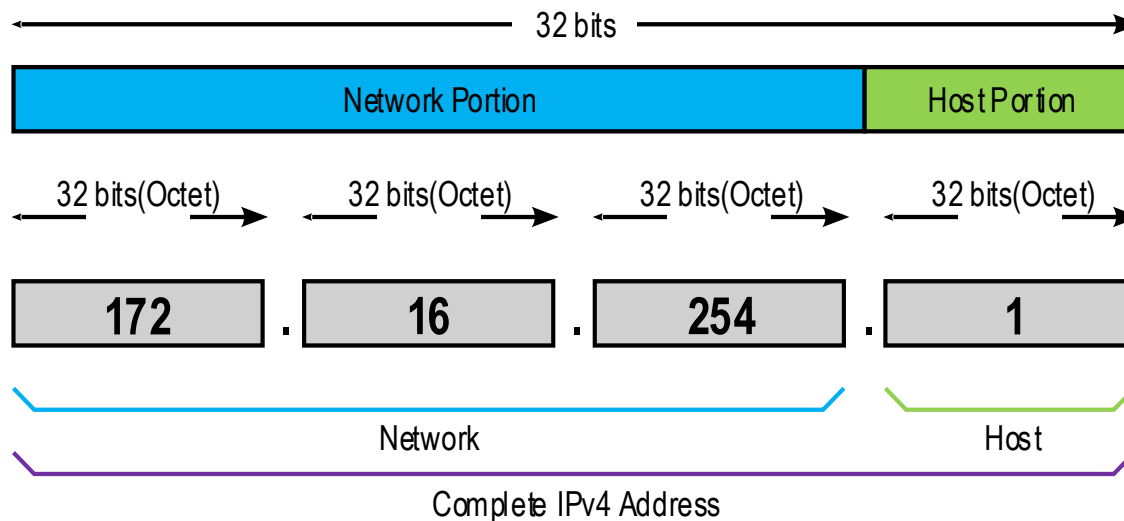




- 48-bits long **MAC address** burned to ROM of NIC
 - Each address should be unique (at least on the segment)
 - Flat address space assigned by IEEE

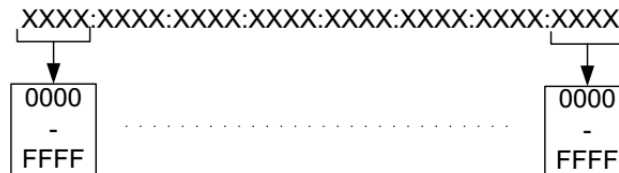


- Address consists of two parts:
network identification (NetId) + host identification (HostId)
 - E.g., 147.229.0.0 as NetId and 0.0.176.14 as HostId

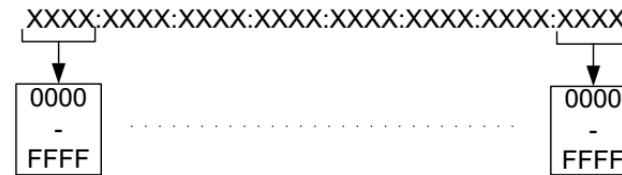


- Where is the border between NetId and HostId?*
 - Classful addressing vs. Classless addressing

- $x:x:x:x:x:x:x:x$, where x is a 16-bit



- RFC 5952 defines rules how to display IPv6 addresses
- Leading zeros **MUST** be suppressed
 - $2001:0db8::0001 \rightarrow 2001:db8::1$
- A single 16-bit 0000 field **MUST** be represented as 0
- $:: \rightarrow$ **Shorten as Much as Possible**
 - When the length of the consecutive 16-bit 0 fields are equal (i.e., $2001:db8:0:0:1:0:0:1$), the first sequence of zero bits **MUST** be shortened.
 - $::$ **MUST NOT** be used to shorten just one 16-bit 0 field
- The characters **MUST** be represented in lowercase.



- hexadecimal field
- Leading zeros in a field are optional:
 - 2031:0:130f:0:0:9c0:876a:130b
- Only once per address successive fields of 0 can be represented as ::

2031:0000:130f:0000:0000:09c0:876a:130b

2031:0:130f::9c0:876a:130b

ff01:0:0:0:0:0:0:1 >>> ff01::1

0:0:0:0:0:0:0:1 >>> ::1

0:0:0:0:0:0:0:0 >>> ::

- *Host knows.*
 - IP address assigned by administrator
- *Host does not know.*
 - MAC address assigned by manufacturer
- Glue between Layer 2 and Layer 3 addresses
 - ARP is for IPv4 to Ethernet MAC resolution
 - ND is for IPv6 to Ethernet MAC resolution
- When PDU are being encapsulated, host can't leave destination MAC address field blank.
- Each IP-to-MAC binding stored in local cache

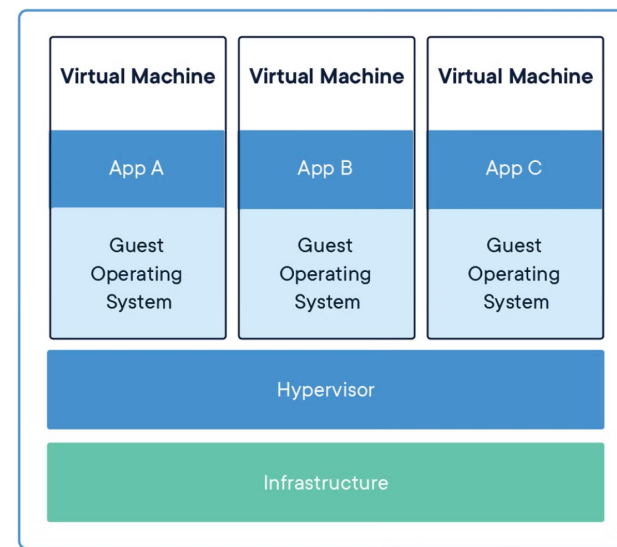
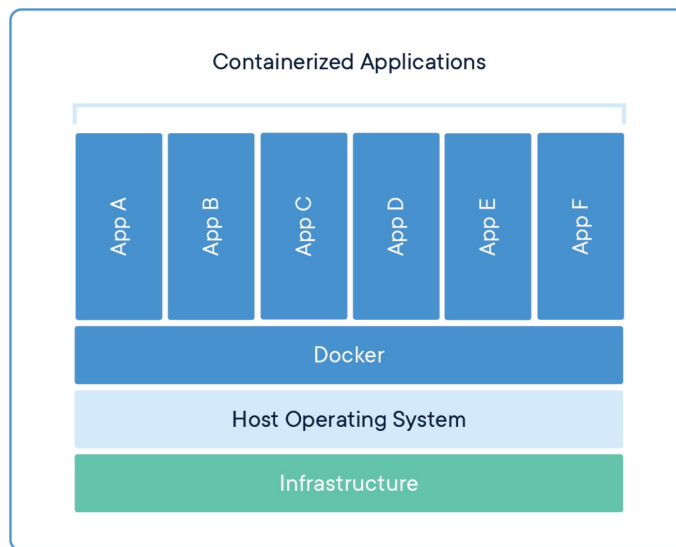
- L2:
 - WiFi: WPA123, WPS, protected management frames
 - Port security, DHCP spoofing/snooping, ARP MitM / Dynamic ARP inspection
- L3: GRE, IPsec, VPN – SSL (openvpn), wireguard
- L4: TLS/SSL, PKI
- L7:
 - DNS – DoH, DoT, DNSSEC
 - Email - DKIM, SPF, DMARC, PGP, S/MIME
 - Authentication: Radius, LDAP, OAuth, JWT

Containers

- Complex software environments
 - OS type and versions, Drivers, Compiler type and versions – Software dependencies
 - Python/Python libraries (virtualenv)
- Reproducibility - package data and code
- Development, Automatization

- Main two categories:
 - Empowering developers with APIs for consuming infrastructure.
 - Software packaging and distribution.
- This has become the standard application design in the devops/cloud native world.
- Microservice architectures being the prime example of this design

- Hardware virtualization
 - Running multiple OSes on the same hardware – VMWare, VirtualBox
- OS level virtualization
 - run isolated OS instance (guest) under a server OS (host)



- Docker
 - Established
 - docker hub for container sharing
- OS vendor tools
 - RedHat Podman
- LXC – mainly supported on Debian
- Orchestration tools
 - use containers to spin up groups of servers –
Kubernetes, Docker Compose

- How do we network our containers together?
- Do we need to share a layer 2 segment between containers on different nodes?
- Do we need IP routing?
- Do we need to provide IP services for containers?

- Play with basic concepts:
 - <https://training.play-with-docker.com/#ops>